

E.3 / PAPER 2 / QUESTION 1

30 April 2025 / TZ3 (file label) / HL Paper 2 / original Q5 / 2225-6721

1. (a) A stationary isotope of $_{88}\text{Ra}$ (Radium) decays into Rn (Radon) and an alpha particle.

(i) Identify the proton number of Rn.

[1]

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The following data are available:

Atomic mass of Ra = 226.02540 u

Atomic mass of Rn = 222.01757 u

Mass of α = 4.00260 u

(ii) Show that the energy released in this decay is about 5 MeV.

[2]

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(iii) Show that about 98 % of the energy released is carried by the alpha particle.

[2]

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(Question 1 continued)

- (b) In an experiment performed by Geiger, Marsden and Rutherford to investigate the atomic structure, alpha particles were directed into a thin metal foil.

(i) Describe two observations of this experiment.

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(ii) Outline how the experiment was interpreted.

[2]

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(iii) In this experiment, $^{197}_{79}\text{Au}$ (gold) foil was used.

Calculate the distance of closest approach for alpha particles of 4.8 MeV in head-on scattering.

[3]

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(Question 1 continued)

(iv) Calculate $\frac{\text{radius of } {}^{197}_{79}\text{Au nucleus}}{\text{radius of } {}^{27}_{13}\text{Al (aluminium) nucleus}}$. [2]

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